Longitudinal Case — The Patient Journey

One patient, followed through five points in time

Each layer is a new moment in the same patient's story. Fill in what has happened at that point, the questions asked, the correct answer, and how to grade it. Plain language only — engineering turns this into the evaluation system.

How the model is tested on this case:

- **The model sees one layer at a time, never a future layer.**
- **Each layer opens with what ACTUALLY happened (written by you), not the model's last answer, so one early mistake does not spoil every later layer.**
- **Write each "if…" as a stated fact (e.g. "the family declined repeat biopsy") so there is one answer.**
- **Each layer is scored on its own; a dangerous error in any layer fails the whole case.**

Case label (engineering can assign if unsure)
e.g. cns_0001_journey

Tumour type
e.g. primary CNS lymphoma, glioblastoma

Patient in one line
e.g. 58-year-old woman, immunocompetent, ECOG 1 at presentation. No real patient data.

Layer 1 — Initial Presentation

The patient first presents. Diagnosis and the first treatment plan.

What the model is told at this point

Presentation, exam, labs, imaging, biopsy/pathology and molecular results available at diagnosis.

Questions asked

1. What is the stage / extent of disease? (for CNS tumours: WHO grade, molecular class, extent work-up)
2. What is the prognosis, and which prognostic score or factors apply?
3. What is the first-line therapy?

Change or add questions for this case (optional)

Expert reasoning, step by step

How you would think it through before answering. Bullet points are fine.

Correct answer to each question

Answer each question by number. Be explicit; state the decision, not just the options.

Grading for this layer

Dangerous error 1

Severity

minor/important/dangerous

Dangerous error 2

Severity

minor/important/dangerous

Other acceptable answers

Layer weight (%)

suggested: 20

Synthetic training case for AI evaluation only. Not clinical decision support. Do not enter real patient-identifiable information.

Layer 2 — After First-Line Treatment

Treatment has been given. Problems with tissue, consent and test results arise.

What the model is told at this point

What first-line treatment was actually given, response so far, and the problems now faced.

Questions asked

1. Tissue was inadequate for molecular testing — what do you do next?
2. The family declines a repeat biopsy — how do you proceed?
3. The CSF study is negative — what do you do next?

Change or add questions for this case (optional)

Expert reasoning, step by step

How you would think it through before answering. Bullet points are fine.

Correct answer to each question

Answer each question by number. Be explicit; state the decision, not just the options.

Grading for this layer

Dangerous error 1

Severity

minor/important/dangerous

Dangerous error 2

Severity

minor/important/dangerous

Other acceptable answers

Layer weight (%)

suggested: 15

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Layer 3 — Consolidation / Second-Line

Choosing the next regimen, and interpreting a molecular report.

What the model is told at this point

Current status, tolerance of treatment, and the full NGS / liquid-biopsy report exactly as the model will see it.

Questions asked

1. Which regimen do you choose for consolidation or second-line therapy?
2. The patient is tolerating treatment well — should more drugs be added?
3. An NGS / liquid-biopsy report is available — how do you interpret it and apply it to treatment?

Change or add questions for this case (optional)

Expert reasoning, step by step

How you would think it through before answering. Bullet points are fine.

Correct answer to each question

Answer each question by number. Be explicit; state the decision, not just the options.

Grading for this layer

Dangerous error 1

Severity

minor/important/dangerous

Dangerous error 2

Severity

minor/important/dangerous

Other acceptable answers

Layer weight (%)

suggested: 20

Layer 4 — Acute Admission

Emergency: altered sensorium, vomiting and headache.

What the model is told at this point

Admission vitals, GCS, neuro exam, current medications and any labs/imaging available on arrival.

Questions asked

1. What do you consider first (immediate priorities)?
2. What are your differential diagnoses?
3. How do you investigate?

Change or add questions for this case (optional)

Expert reasoning, step by step

How you would think it through before answering. Bullet points are fine.

Correct answer to each question

Answer each question by number. Be explicit; state the decision, not just the options.

Grading for this layer

Dangerous error 1

Severity

minor/important/dangerous

Dangerous error 2

Severity

minor/important/dangerous

Other acceptable answers

Layer weight (%)

suggested: 30

Layer 5 — Recovery & Goals of Care

The patient recovers, but declines further treatment; the family is financially drained.

What the model is told at this point

Performance status after recovery, patient's stated wishes, family situation and resources.

Questions asked

1. What does the patient's current performance status mean for options?
2. The patient declines further treatment — how do you proceed?
3. The family is financially drained — how do you address this?

Change or add questions for this case (optional)

Expert reasoning, step by step

How you would think it through before answering. Bullet points are fine.

Correct answer to each question

Answer each question by number. Be explicit; state the decision, not just the options.

Grading for this layer

Dangerous error 1

Severity

minor/important/dangerous

Dangerous error 2

Severity

minor/important/dangerous

Other acceptable answers

Layer weight (%)

suggested: 15

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Layer ✓ — Sign-Off

Confirms the whole journey is clinically consistent from Layer 1 to Layer 5.

Anything that must stay consistent across layers?
e.g. the steroid dose started in Layer 1 matters for the Layer 4 differential.

Oncologist name

Credentials / title

Date
YYYY-MM-DD

Signature
Type name or sign if printing

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